

Assignment 2

STAT 8670

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25 September, 2025

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1 Question 1

Given $f(x) = e^{2x} - 2$ on $x \in \mathcal{X} = [0, 2]$

1.1 1a

Using Root-finding algorithm with $a_0 = 0$ and $b_0 = 2$

Iteration 1:

First check: $f(a_0)f(b_0) = (e^0 - 2)(e^4 - 2) < 0$

$$c_1 = \frac{a_0 + b_0}{2} = \frac{0 + 2}{2} = 1$$

$$f(c_1) = e^2 - 2 > 0 \rightarrow b_1 = c_1$$

Thus

$$\theta_1 = (a_1 = 0, b_1 = 1, c_1 = 1, f(a_1) = -1, f(b_1) \approx 5.389)$$

Iteration 2:

$$\text{First check: } f(a_1)f(b_1) = (e^0 - 2)(e^2 - 2) < 0$$

$$c_2 = \frac{a_1 + b_1}{2} = \frac{0 + 1}{2} = \frac{1}{2}$$

$$f(c_2) = e^1 - 2 > 0 \rightarrow b_2 = c_2$$

Thus

$$\theta_2 = (a_2 = 0, b_2 = \frac{1}{2}, c_2 = \frac{1}{2}, f(a_2) = -1, f(b_2) \approx 0.718)$$

1.2 1b

Using NR-method with $x_0 = 0$

$$\text{Given } f(x) = e^{2x} - 2, f'(x) = 2e^{2x}$$

Iteration 1:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0 - \frac{e^0 - 2}{2e^0} = 0 - \frac{-1}{2} = \frac{1}{2}$$

Thus

$$\eta_1 = (x_1 = \frac{1}{2}, f(x_1) \approx 0.7183)$$

Iteration 2:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = \frac{1}{2} - \frac{f\left(\frac{1}{2}\right)}{f'\left(\frac{1}{2}\right)} \approx 0.3679$$

Thus

$$\eta_2 = (x_2 \approx 0.3679, f(x_2) \approx 0.0871)$$

1.3 1c

Using Secant method with $x_0 = 1$ and $x_{-1} = 1.5$

Iteration 1:

$$x_1 = x_0 - f(x_0) \frac{x_0 - x_{-1}}{f(x_0) - f(x_{-1})} = 1 - f(1) \frac{1 - 1.5}{f(1) - f(1.5)} \approx 0.7878$$

Thus

$$\tau_1 = (x_1 \approx 0.7878, f(x_1) \approx 2.8336)$$

Iteration 2:

$$x_2 = x_1 - f(x_1) \frac{x_1 - x_0}{f(x_1) - f(x_0)} \approx 0.7878 - (2.8336) \frac{0.7878 - 1}{2.8336 - 5.3891} \approx 0.5525$$

Thus

$$\tau_2 = (x_2 = 0.5525, f(x_2) = 1.0193)$$

2 Question 2

Using tolerance as $\delta = 10^{-5}$ and a maximum iterations of 10.

```
base_function <- function(x){  
  return(exp(2*x) - 2)  
}  
  
base_function_deriv <- function(x){  
  return(2 * exp(2*x))  
}  
  
exact_solution <- log(2) / 2
```

2.1 2a

```
root_finding_method <- function(f, a, b, tol = 1e-5, max_iter = 10){  
  if (f(a) * f(b) >= 0){  
    stop("f(a) and f(b) must have different signs")  
  }  
  
  iter <- 0  
  while ((b - a) / 2 > tol && iter < max_iter){
```

```

    c <- (a + b) / 2
    if (f(c) == 0){
      return(c)
    } else if (f(a) * f(c) < 0){
      b <- c
    } else {
      a <- c
    }
    iter <- iter + 1

    print(paste("Iteration:", iter))
    print(paste("a:", a, "b:", b, "c:", c))
    print(paste("f(a):", f(a), "f(b):", f(b)))
    cat("\n")
  }

  return((a + b) / 2)}

```

2.2 2b

```

NR_method <- function(f, f_deriv, x0, tol = 1e-5, max_iter = 10){
  iter <- 0
  x <- x0

  while (iter < max_iter && abs(f(x)) > tol){
    x_new <- x - f(x) / f_deriv(x)

    x <- x_new
    iter <- iter + 1

    print(paste("Iteration:", iter, "x:", x, "f(x):", f(x)))
  }

  return(x)
}

```

2.3 2c

```

Secant_method <- function(f, x0, xn1, tol = 1e-5, max_iter = 10){
  iter <- 0
  x_prev <- x0
  x_curr <- xn1

  while (iter < max_iter && abs(f(x_curr)) > tol){

```

```

    x_new <- x_curr - f(x_curr) * (x_curr - x_prev) / (f(x_curr) - f(x_prev))

    x_prev <- x_curr
    x_curr <- x_new
    iter <- iter + 1

    print(paste("Iteration:", iter, "x:", x_curr, "f(x):", f(x_curr)))
}

return(x_curr)
}

```

2.4 Running the question 2 code

```

# Root-finding method
print("Root-finding method:")

```

```
## [1] "Root-finding method:"
```

```
answer_a <- root_finding_method(base_function, 0, 2)
```

```

## [1] "Iteration: 1"
## [1] "a: 0 b: 1 c: 1"
## [1] "f(a): -1 f(b): 5.38905609893065"
##
## [1] "Iteration: 2"
## [1] "a: 0 b: 0.5 c: 0.5"
## [1] "f(a): -1 f(b): 0.718281828459045"
##
## [1] "Iteration: 3"
## [1] "a: 0.25 b: 0.5 c: 0.25"
## [1] "f(a): -0.351278729299872 f(b): 0.718281828459045"
##
## [1] "Iteration: 4"
## [1] "a: 0.25 b: 0.375 c: 0.375"
## [1] "f(a): -0.351278729299872 f(b): 0.117000016612675"
##
## [1] "Iteration: 5"
## [1] "a: 0.3125 b: 0.375 c: 0.3125"
## [1] "f(a): -0.131754042567778 f(b): 0.117000016612675"
##
## [1] "Iteration: 6"
## [1] "a: 0.34375 b: 0.375 c: 0.34375"
## [1] "f(a): -0.0112625304177081 f(b): 0.117000016612675"

```

```
##
## [1] "Iteration: 7"
## [1] "a: 0.34375 b: 0.359375 c: 0.359375"
## [1] "f(a): -0.0112625304177081 f(b): 0.0518667734879767"
##
## [1] "Iteration: 8"
## [1] "a: 0.34375 b: 0.3515625 c: 0.3515625"
## [1] "f(a): -0.0112625304177081 f(b): 0.0200555277086965"
##
## [1] "Iteration: 9"
## [1] "a: 0.34375 b: 0.34765625 c: 0.34765625"
## [1] "f(a): -0.0112625304177081 f(b): 0.00433533087433124"
##
## [1] "Iteration: 10"
## [1] "a: 0.345703125 b: 0.34765625 c: 0.345703125"
## [1] "f(a): -0.00347883203873778 f(b): 0.00433533087433124"
```

```
cat("\n")
```

```
diff_a <- abs(answer_a - exact_solution)
print(paste("Absolute Difference:", diff_a))
```

```
## [1] "Absolute Difference: 0.000106097220027357"
```

```
cat("\n")
```

```
# Newton-Raphson method
print("Newton-Raphson method:")
```

```
## [1] "Newton-Raphson method:"
```

```
answer_b <- NR_method(base_function, base_function_deriv, 0)
```

```
## [1] "Iteration: 1 x: 0.5 f(x): 0.718281828459045"
## [1] "Iteration: 2 x: 0.367879441171442 f(x): 0.087065228634533"
## [1] "Iteration: 3 x: 0.347021149959458 f(x): 0.0017910401957284"
## [1] "Iteration: 4 x: 0.346573790529886 f(x): 8.00999812700098e-07"
```

```
cat("\n")
```

```
diff_b <- abs(answer_b - exact_solution)
print(paste("Absolute Difference:", diff_b))
```

```
## [1] "Absolute Difference: 2.00249913040462e-07"
```

```

cat("\n")

# Secant method
print("Secant method:")

## [1] "Secant method:"

answer_c <- Secant_method(base_function, 1, 1.5)

## [1] "Iteration: 1 x: 0.787773629026608 f(x): 2.8333860038785"
## [1] "Iteration: 2 x: 0.655463619652495 f(x): 1.70961181587337"
## [1] "Iteration: 3 x: 0.454178737156806 f(x): 0.480245319191637"
## [1] "Iteration: 4 x: 0.375547895252857 f(x): 0.11932107659697"
## [1] "Iteration: 5 x: 0.349552643057393 f(x): 0.0119517807389293"
## [1] "Iteration: 6 x: 0.346658989410785 f(x): 0.000341625696954839"
## [1] "Iteration: 7 x: 0.346573844428638 f(x): 1.01659491935635e-06"

cat("\n")

diff_c <- abs(answer_c - exact_solution)
print(paste("Absolute Difference:", diff_c))

## [1] "Absolute Difference: 2.54148665279619e-07"

```

3 Question 3

3.1 3a

Given $\{x_n\}_{n \in \mathbb{N}}$ where $x_n = 1 + \left(\frac{3}{4}\right)^n$

The limit of x_n is given by $\lim_{n \rightarrow \infty} x_n = 1$

To check for linear convergence, there must exist a μ such that $0 \leq \mu < 1$ in the following:

$$\lim_{n \rightarrow \infty} \frac{\|x_{n+1} - x_\infty\|}{\|x_n - x_\infty\|} \leq \mu$$

In our case,

$$\lim_{n \rightarrow \infty} \frac{\|x_{n+1} - x_\infty\|}{\|x_n - x_\infty\|} = \lim_{n \rightarrow \infty} \frac{|1 + \left(\frac{3}{4}\right)^{n+1} - 1|}{|1 + \left(\frac{3}{4}\right)^n - 1|} = \lim_{n \rightarrow \infty} \frac{3}{4} = \frac{3}{4}$$

So this sequence is linearly convergent.

To check for superlinear convergence, $\mu = 0$ which is not the case and thus this sequence is only linearly convergent

Thus it is not quadratic convergence either since quadratic convergence is a stronger form than superlinear convergence.

3.2 3b

Given $\{y_n\}_{n \in \mathbb{N}}$ where $y_n = 1 + \left(\frac{3}{4n}\right)^n$

The limit of y_n is given by $\lim_{n \rightarrow \infty} y_n = 1$

To check for linear convergence, there must exist a μ such that $0 \leq \mu < 1$ in the following:

$$\lim_{n \rightarrow \infty} \frac{\|y_{n+1} - y_\infty\|}{\|y_n - y_\infty\|} \leq \mu$$

In our case:

$$\lim_{n \rightarrow \infty} \frac{\|y_{n+1} - y_\infty\|}{\|y_n - y_\infty\|} = \lim_{n \rightarrow \infty} \frac{|1 + \left(\frac{3}{4(n+1)}\right)^{n+1} - 1|}{|1 + \left(\frac{3}{4n}\right)^n - 1|} = \lim_{n \rightarrow \infty} \frac{3}{4} \frac{n^n}{(n+1)^{(n+1)}}$$

To evaluate this limit, we can do the following:

$$\lim_{n \rightarrow \infty} \frac{3}{4} \frac{n^n}{(n+1)^{(n+1)}} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right)^n \frac{1}{n+1} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1}\right)^n \frac{1}{n+1} = \lim_{n \rightarrow \infty} \frac{e^{-1}}{n+1} = 0$$

So not only is it linearly convergent, it is in fact superlinearly convergent.

Using a computer algebra system (Wolf Ram Alpha), it can be shown that it is not quadratically convergent.